

BUS 4101

Rapid Revision Guide

E-Commerce and Management Information Systems

Built for one-night preparation:

- Starred suggestions first, then all remaining suggestions.
- Concise exam-ready notes in easy English.
- Correct formulas and fully worked calculations.
- Definition bank, difference bank, mnemonic bank and keyword bank.
- Lecture-only rescue topics selected from the class notes and the 2024 question pattern.

Department of Computer Science and Engineering
Jashore University of Science and Technology

Semester Final Examination Preparation

Source basis: BUS 4101 suggestion sheet, Classes 01–39 notes, and the supplied 2024 examination paper. Topics marked as supplements are concise additions for items not fully explained in the notes.

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1 Fastest Preparation Strategy

STAR PRIORITY

Study these seven double-starred topics before anything else:

1. Critical Path Method (CPM)
2. Porter's Value Chain Model
3. AIDA Funnel Model
4. Economic Order Quantity (EOQ)
5. RFM Customer Segmentation
6. Kotter's 8-Step Change Model
7. Expected Value Mathematics

1.1 Recommended order for tonight

Block	What to study	Target time
1	Seven starred topics, including all worked mathematics	2 h 30 min
2	E-commerce core: architecture, models, metrics, payment, fulfilment, MFS fraud	1 h 30 min
3	MIS core: IS hierarchy, ERP, DIKW, decision models, EIS/GDSS, CIA	1 h 40 min
4	Definition and difference banks	45 min
5	Lecture-only rescue topics and predicted questions	45 min
6	Formula sheet + interactive quiz/math lab	50 min

Exam Tip

The supplied 2024 paper used eight sets and required answering any six. Questions mixed short definitions, comparisons, explanations, and application-based analysis. Therefore, prepare at least six broad clusters rather than memorizing isolated definitions.

1.2 How to write for marks

- **1 mark:** one exact definition in 2–3 lines.
- **3 marks:** definition + 3 strong points or a small comparison table.
- **4–5 marks:** definition + labelled diagram/model + 4–5 explanations + example.
- **6–8 marks:** structured answer with introduction, framework, application/case, advantages or implications, and conclusion.
- For mathematics, always write: **formula, variable meanings, substitution, calculation, final decision/interpretation.**

2 Starred High-Priority Topics

2.1 Critical Path Method (CPM) HIGH PRIORITY

Supplement added because the suggestion topic was not fully covered in the notes

The suggestion sheet marks CPM as important, but the supplied class notes only mention critical-path scheduling briefly. The following is a standard, exam-ready CPM treatment.

Definition

The **Critical Path Method (CPM)** is a project scheduling technique that identifies the longest-duration path through a network of dependent activities. Activities on this path have **zero total float**; delaying any of them delays the whole project.

Formula / Method

For activity i of duration d_i :

$$EF_i = ES_i + d_i, \quad LS_i = LF_i - d_i$$

$$ES_i = \max(EF \text{ of predecessors}), \quad LF_i = \min(LS \text{ of successors})$$

$$\text{Total Float} = LS - ES = LF - EF$$

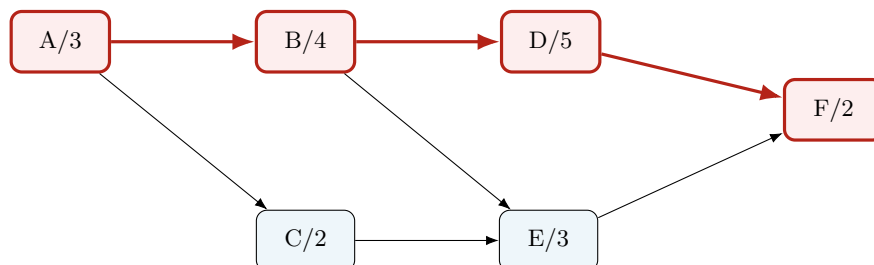
An activity is critical when its total float is zero.

Procedure mnemonic: F-B-F-C

1. Forward pass: calculate ES and EF.
2. Backward pass: calculate LF and LS.
3. Float: calculate LS-ES.
4. Critical path: connect zero-float activities.

Worked example

Activity	Predecessor	Duration (days)
A	–	3
B	A	4
C	A	2
D	B	5
E	B,C	3
F	D,E	2



Act.	d	ES	EF	LS	LF	Float	Status
A	3	0	3	0	3	0	Critical
B	4	3	7	3	7	0	Critical
C	2	3	5	7	9	4	Non-critical
D	5	7	12	7	12	0	Critical
E	3	7	10	9	12	2	Non-critical
F	2	12	14	12	14	0	Critical

Answer: Critical path $A \rightarrow B \rightarrow D \rightarrow F$; project duration = $3 + 4 + 5 + 2 = 14$ days.

Exam Tip

At a merge node, use the **maximum** predecessor EF in the forward pass. At a split node, use the **minimum** successor LS in the backward pass. These are the two most common errors.

2.2 Porter's Value Chain Model **HIGH PRIORITY**

Definition

Porter's Value Chain analyzes the activities a firm performs to create and deliver value. Each activity either adds value or adds cost. Competitive advantage comes from performing activities more efficiently or more distinctively than competitors.

Procurement				
Technology Development				
Human Resource Management				
Firm Infrastructure				
Inbound Logistics	Operations	Outbound Logistics	Marketing & Sales	Service

Primary activities:

- **Inbound logistics:** supplier portals, inventory receiving, RFID/barcodes.
- **Operations:** order processing, warehouse management, automated picking.
- **Outbound logistics:** fulfilment, tracking, last-mile delivery, returns.
- **Marketing and sales:** SEO, ads, email, social media, recommendations.
- **Service:** chatbots, return portals, reviews, warranty and loyalty.

Support activities: firm infrastructure, HR management, technology development, and procurement.

Application: Daraz. Seller onboarding and inbound inventory feed operations; warehouse systems process orders; courier integration performs outbound logistics; campaigns and recommendations drive sales; return/refund support creates service value. ERP, cloud, HR and procurement enable all primary activities.

Mnemonic: I Operate Outlets, Market, Serve = Inbound, Operations, Outbound, Marketing, Service. Support: **Firm Humans Use Procurement Technology** = Infrastructure, HR, Procurement, Technology.

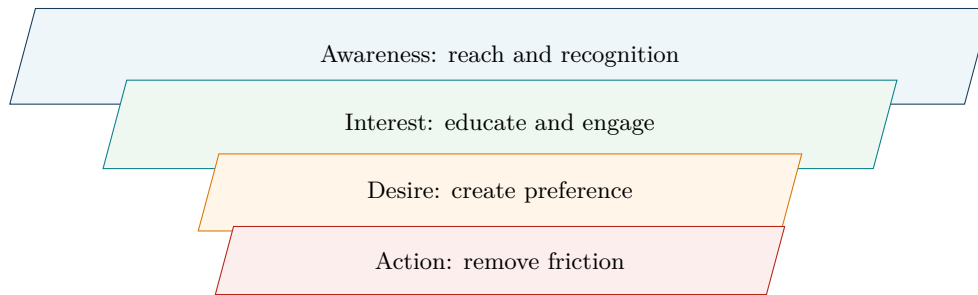
Exam Tip

Do not confuse a firm's value chain with the entire industry supply chain. A value chain describes activities **inside one firm**; an industry chain links raw-material producers, manufacturers, wholesalers, retailers and consumers.

2.3 AIDA Funnel Model **HIGH PRIORITY**

Definition

AIDA describes the four stages through which a customer moves before purchase: **Awareness, Interest, Desire, Action**. Each stage requires different marketing tactics and metrics.



Stage	Customer state / objective	Digital tactics	Main metrics
Awareness	Does not know the brand; create recognition	SEO, broad social ads, display ads, influencer content, PR	Reach, impressions, new visitors
Interest	Evaluates options; educate and engage	Buying guides, videos, newsletters, retargeting, demos	Time on site, pages/session, sign-ups
Desire	Wants product but lacks conviction	Reviews, social proof, scarcity, personalization, live chat	Add-to-cart, wish-list, product dwell time
Action	Ready to buy; remove purchase friction	Guest checkout, multiple payments, trust badges, clear delivery, strong CTA	Conversion, abandonment, revenue, AOV

Application to digital accessories: show a short influencer video to create awareness; publish comparison guides for interest; use verified reviews and bundle discounts for desire; provide bKash/card/COD and a one-page checkout for action.

Mnemonic: **Attract - Inform - Develop Want - Ask for Order.**

2.4 Inventory Management and EOQ **HIGH PRIORITY**

Definition

Inventory management is the process of ordering, storing, tracking and controlling stock. EOQ is the order size that minimizes the sum of annual ordering cost and annual holding cost.

Formula / Method

$$Q^* = EOQ = \sqrt{\frac{2DS}{H}}$$

D = annual demand; S = ordering cost per order; H = annual holding cost per unit.

$$TC(Q) = \frac{D}{Q}S + \frac{Q}{2}H$$

$$\text{Reorder Point} = \text{Daily Demand} \times \text{Lead Time} + \text{Safety Stock}$$

Worked example from the class-note pattern: $D = 1,200$ units/year, $S = 500$ BDT/order, $H = 50$ BDT/unit/year.

$$EOQ = \sqrt{\frac{2(1200)(500)}{50}} = \sqrt{24,000} = 154.92 \approx 155 \text{ units}$$

Orders per year = $1200/155 \approx 7.74$, so place about 8 orders/year.

At $Q = 155$:

$$\text{Ordering cost} \approx \frac{1200}{155}(500) = 3,871 \text{ BDT}$$

$$\text{Holding cost} \approx \frac{155}{2}(50) = 3,875 \text{ BDT}$$

The two costs are approximately equal at EOQ, which is a useful accuracy check.

Assumptions: constant known demand, constant lead time and costs, no quantity discounts, no stockouts, immediate replenishment. Real e-commerce demand may violate these assumptions.

Mnemonic: **Double Demand and Setup, divide by Holding, then Root.**

2.5 RFM Model for Market Strategy **HIGH PRIORITY**

Definition

RFM is a customer-segmentation framework based on **Recency** (how recently a customer purchased), **Frequency** (how often), and **Monetary value** (how much was spent). Each dimension is commonly scored 1–5, with 5 best.

Formula / Method

$$\text{RFM Total Score} = R + F + M, \quad 3 \leq R + F + M \leq 15$$

A three-digit code such as 515 retains more information than only the sum: it means very recent, low frequency, high spending.

Segment	Typical score	Meaning	Suitable strategy
Champions	555	Recent, frequent, highest value	VIP rewards, early access, referrals
Loyal	mostly 3–5	Regular and valuable	Upsell, membership, bundles
Potential loyalists	high R, medium/low F, M	New but promising	Onboarding and second-purchase reward
Promising	high R, low F	Recent first buyer	Education and a second-order incentive
At risk	low R, high F, M	Used to buy regularly	Personalized win-back offer
Cannot lose	R=1, F/M=4–5	Former best customer now inactive	Personal outreach, premium recovery offer
Hibernating	low all	Old, rare, low spend	Low-cost reactivation only
Lost	111	Lowest value and inactive	Minimal spend or list clean-up

Mini case: Customer X bought 2 days ago, placed 12 orders, spent 45,000 BDT: likely 555, so treat as Champion. Customer Y last bought 8 months ago but previously ordered 15 times and spent heavily: likely 155, so “Cannot Lose Them”; use urgent personal win-back.

Mnemonic: **Recent, Regular, Revenue.**

2.6 Kotter’s 8-Step Change Model **HIGH PRIORITY**

Definition

Kotter’s model is a structured method for leading organizational change. It is especially relevant to ERP/MIS implementation because many failures arise from employee resistance rather than technology.

1. **Create urgency:** show why the current state is unsustainable.
2. **Build a guiding coalition:** unite respected leaders from major departments.
3. **Form a strategic vision:** define measurable change outcomes.
4. **Enlist a volunteer army:** communicate and recruit advocates.
5. **Enable action:** remove barriers, train users and redesign workflows.
6. **Generate short-term wins:** publicize early measurable success.
7. **Sustain acceleration:** expand and improve; do not declare victory early.
8. **Institute change:** embed the new system in SOPs, jobs and culture.

ERP application: Urgency may be delayed closing and inaccurate inventory; the coalition includes finance, production, sales and IT; the vision sets targets; pilot inventory produces a quick win; later modules are rolled out; use becomes part of performance evaluation.

Mnemonic sentence: **Urgent Coalitions Visualize Volunteers; Enable Wins, Sustain, Institute.**

2.7 Expected Value Mathematics **HIGH PRIORITY**

Definition

Expected Value (EV) is the probability-weighted average payoff of an alternative. It supports decisions under risk, where possible outcomes and their probabilities are known.

Formula / Method

$$EV = \sum_{i=1}^n P_i V_i, \quad \sum P_i = 1$$

For a decision tree, calculate EV at each chance node and roll back from right to left; select the highest EV at each decision node.

Worked example:

- Product A: high demand 0.50×6.0 million; medium 0.30×2.5 million; low $0.20 \times (-1.0)$ million.
- $EV(A) = 3.0 + 0.75 - 0.20 = 3.55$ million BDT.
- Product B: high 0.30×8.0 ; medium 0.40×3.0 ; low $0.30 \times (-2.0)$ million.
- $EV(B) = 2.4 + 1.2 - 0.6 = 3.00$ million BDT.

Decision: choose Product A because $3.55 > 3.00$ million BDT.

Exam Tip

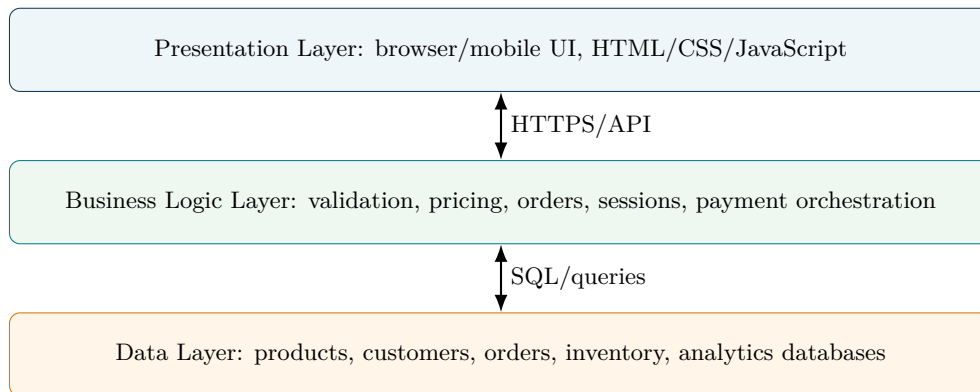
Probabilities must sum to 1. EV maximization is rational over repeated decisions, but it does not capture risk tolerance, worst-case survival, ethics or strategic fit.

3 Remaining Suggested E-Commerce Topics

3.1 E-Commerce Architecture

Definition

A multi-tier e-commerce architecture separates an application into layers with distinct responsibilities, improving scalability, maintainability and security.



Why tiers? independent scaling, easier maintenance, parallel development, reuse and database isolation. At large scale, add CDN, load balancer, web servers, caches, read replicas and horizontal auto-scaling.

3.2 E-Commerce Business Models and E-Commerce vs F-Commerce

Business model explains who the participants are, what value is offered and how the firm operates. Common participant models:

- **B2C:** business sells to consumers (Daraz retail, Chaldal).
- **B2B:** business sells to another business (Alibaba wholesale, SaaS).
- **C2C:** consumers transact through a platform (Facebook Marketplace, eBay).
- **C2B:** individuals offer value to businesses (freelancers, influencers).
- **B2G/G2C:** electronic procurement or public services.

Supplement added because the suggestion topic was not fully covered in the notes

F-Commerce means commerce conducted mainly through Facebook pages, groups, Marketplace, Messenger and live selling. It is a subset of social commerce and of e-commerce.

Dimension	E-Commerce	F-Commerce
Platform	Any website/app/marketplace	Facebook ecosystem mainly
Control	Merchant may control own site and data	Depends strongly on Facebook policies/algorithm
Features	Full catalog, checkout, analytics, integrations	Posts, comments, messages, live video, social proof
Entry cost	Can require development/platform cost	Very low and fast to start
Trust	Brand, reviews, payment and policy systems	Social identity, followers, comments, personal communication
Risk	Technical/security/logistics risks	Fake pages, weak records, algorithm dependence, informal transactions

3.3 Revenue Models and Examples

Model	How income is earned	Examples
Advertising	Sell audience attention or ad placement	Google, Facebook, YouTube
Subscription	Recurring fee for continued access	Netflix, Spotify Premium, SaaS
Transaction fee	Percentage/fixed fee per facilitated transaction	Daraz marketplace, Airbnb, Upwork
Freemium	Free basic tier; paid advanced tier	Zoom, Dropbox, LinkedIn
Affiliate	Commission for tracked referral leading to sale	Amazon Associates, Daraz affiliate
Sales/E-tail	Buy or produce goods and retain retail margin	Chaldal, direct-to-consumer brands

Multi-revenue model: a platform can combine several streams. Amazon combines sales, marketplace fees, subscription, advertising, cloud and affiliate income.

3.4 Subscription vs Freemium

Dimension	Subscription	Freemium
Access	Payment normally required for service	Basic service free; advanced service paid
Objective	Predictable recurring revenue	Rapid user acquisition and later conversion
Revenue base	Most active users pay	Small paid percentage funds a large free base
Main risk	Churn and cancellation	Low conversion and high free-user cost
Best metric	MRR, churn, retention	Free-to-paid conversion, activation, CAC
Example	Netflix	Spotify free vs Premium

3.5 E-Commerce Metrics and Formulas

Formula / Method

$$\begin{aligned}
 CR &= \frac{\text{Purchases}}{\text{Visits}} \times 100\% \\
 CPC &= \frac{\text{Ad Spend}}{\text{Clicks}} \\
 CAC &= \frac{\text{Marketing Spend}}{\text{New Customers}} \\
 AOV &= \frac{\text{Revenue}}{\text{Orders}} \\
 CLV &= AOV \times \text{Purchase Frequency} \times \text{Customer Lifespan} \\
 ROAS &= \frac{\text{Revenue from Ads}}{\text{Ad Spend}} \\
 \text{Bounce Rate} &= \frac{\text{Single-page Visits}}{\text{Total Visits}} \times 100\% \\
 \text{Cart Abandonment} &= \left(1 - \frac{\text{Orders}}{\text{Cart Additions}} \right) \times 100\%
 \end{aligned}$$

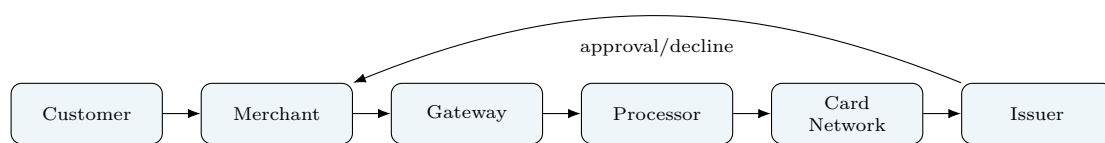
Example: marketing spend 50,000 BDT, 100 new customers, AOV 1,500, four orders/year, lifespan two years. $CAC = 500$ BDT; $CLV = 1,500 \times 4 \times 2 = 12,000$ BDT; $CLV : CAC = 24 : 1$.

3.6 E-Commerce Fulfilment Models

Model	Meaning	Main benefit	Main risk
In-house	Merchant owns/leases warehouse and controls fulfilment	Maximum control and branding	High fixed cost and inventory capital
Dropshipping	Supplier ships directly; merchant holds no stock	Very low entry capital	Low control, slower delivery, supplier errors
3PL	Specialist stores, picks, packs, ships and handles returns	Expertise and economies of scale	Fees and reduced direct control
4PL	Integrator manages the whole network of 3PLs/carriers	End-to-end optimization	Complex and suited to large firms

3.7 Credit/Debit Card Processing Flow

Participants: cardholder, merchant, payment gateway, processor, card network, issuing bank and acquiring bank.



Phase 1 - Authorization (seconds): customer enters data; gateway encrypts; processor routes; network sends to issuer; issuer checks funds, card and fraud score; response returns. Authorization reserves funds, not final transfer.

Phase 2 - Clearing and settlement (batch): merchant submits captured transactions; network calculates obligation/s/interchange; issuer transfers funds through network/acquirer; acquirer credits merchant minus fees.

Gateway vs processor vs acquirer: gateway is the secure technical interface; processor routes messages; acquirer is the merchant's bank and settles funds.

3.8 Cyber Fraud in Mobile Financial Services (MFS)

- **Vishing:** fake bKash/Nagad support call asks for PIN or OTP.
- **Smishing/phishing:** fake SMS/link for prize, refund, KYC or blocked account.
- **SIM-swap/account takeover:** attacker obtains control of phone number and recovery channel.
- **Fake payment screenshot/SMS:** merchant releases goods without verifying account balance/transaction ID.
- **Agent fraud:** unauthorized fees, fake cash-out or misuse of customer credentials.
- **Social-media impersonation:** cloned page/profile requests advance payment.
- **Malware/remote-access app:** victim is persuaded to install software that reads messages or controls device.

Controls: never share PIN/OTP; verify through official app and transaction ID; device binding, MFA, limits and anomaly detection; customer education; agent monitoring; rapid account freeze and incident reporting.

3.9 Network Effects, Winner-Takes-All and Metcalfe's Law

Definition

A network effect exists when a product or platform becomes more valuable as more users join. Direct effects occur among the same user group; indirect effects connect different groups such as buyers and sellers; data effects improve the service through more usage data.

Formula / Method

Metcalfe's Law:

$$V \propto n^2$$

A more exact count of possible pairwise links is $n(n-1)/2$. If users double from n to $2n$, approximate value rises fourfold.

Winner-takes-all: strong network effects, low marginal cost, data advantage, brand and switching costs create a tipping point where one or two platforms capture most value. It weakens when users can multi-home, niches differ greatly, regulation intervenes or local needs remain important.

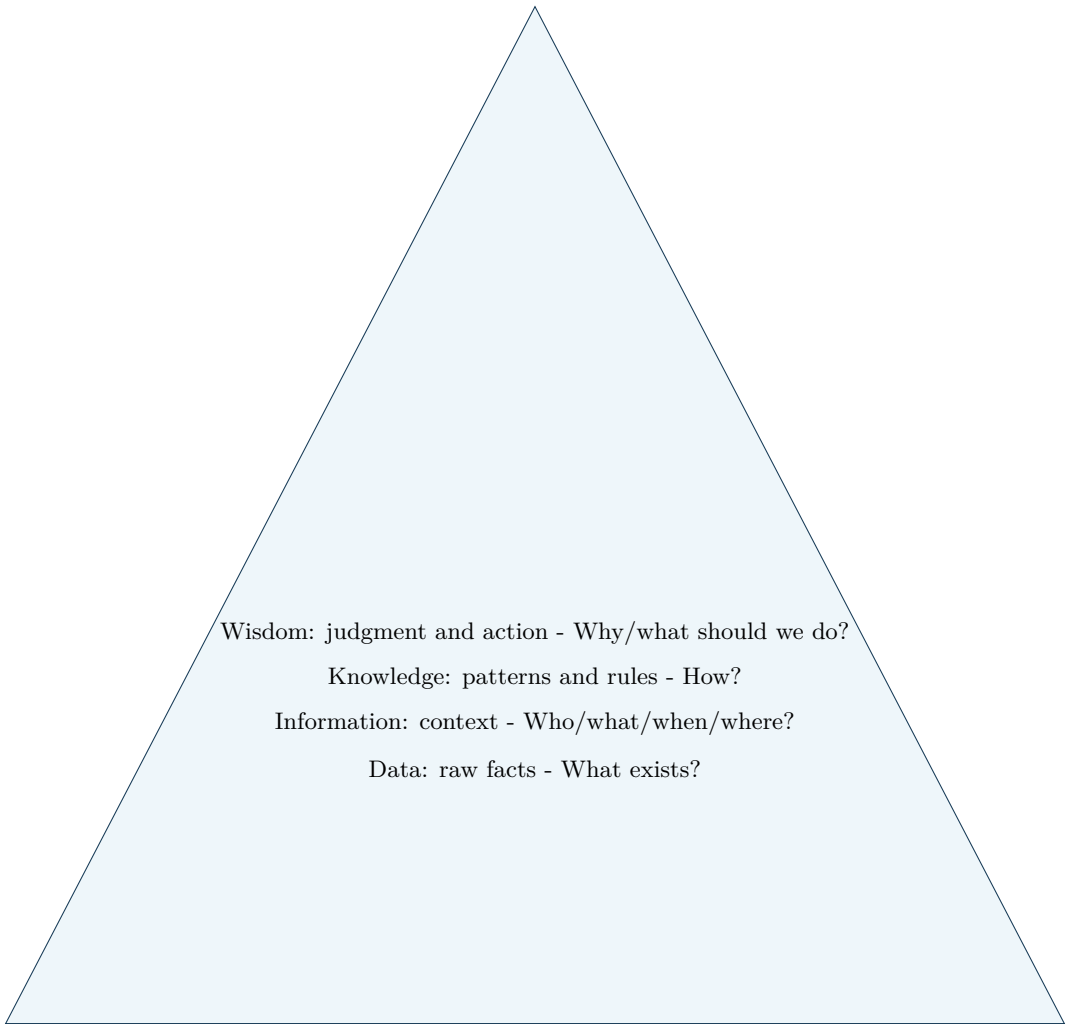
4 Remaining Suggested MIS Topics

4.1 TPS, DPS, MRS, MIS, DSS and EIS

System	Main users	Decision	Output	Core idea/example
DPS	Clerical/operations	Structured	Records, ledgers	Processes data, e.g., payroll or invoice calculation.
TPS	Frontline/operations	Structured	Real-time/batch transactions	Captures reliable high-volume events, e.g., ATM, order entry.
MRS	Middle managers	Structured control	Scheduled, exception, demand reports	Summarizes TPS data into management reports.
MIS	Middle management	Mainly structured	Integrated summaries, dashboards	Provides information for planning, monitoring and control.
DSS	Middle/senior managers	Semi-structured	What-if, models, forecasts	Interactive analysis; supports judgment but does not replace it.
EIS/ESS	Top executives	Unstructured/strategic	Visual KPI dashboard, trends, exceptions	Fast high-level internal and external strategic information.

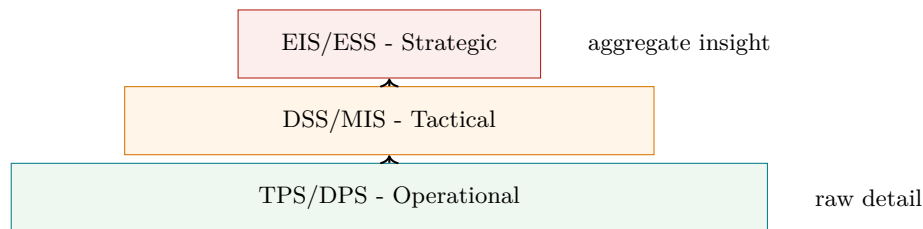
Memory chain: Transactions become Reports, Reports feed Models, Models become Executive Insight.

4.2 DIKW Hierarchy



Example: 4500, 3825, 4100 are data; “sales fell 15%” is information; understanding seasonal decline is knowledge; choosing a targeted promotion without destructive discounting is wisdom.

4.3 Information System Hierarchy



Data flows upward through aggregation; decisions and policies flow downward. Operational systems need detailed real-time internal data; executives need summarized, external and future-oriented information.

4.4 Enterprise Resource Planning (ERP)

Definition

ERP is an integrated information-system platform that connects finance, HR, manufacturing, sales, procurement and supply chain through a unified database and processes. It eliminates departmental information silos and creates one source of truth.

Core modules: finance/accounting, manufacturing/MRP, sales and distribution, procurement, inventory/supply chain, HR/payroll.

Benefits: real-time integration, fewer duplicate entries, faster closing, better planning, process standardization, traceability and management visibility.

Risks: very high cost/time, process disruption, customization complexity, poor data migration, resistance, vendor dependence and implementation failure.

4.5 ERP Implementation in a Manufacturing Firm

Exam case structure:

1. **Problem:** isolated systems, delayed inventory, manual reconciliation, over-promising.
2. **Preparation:** process mapping, feasibility, business case, steering committee and clean master data.
3. **Design/configuration:** finance, sales, inventory, production, procurement and HR modules.
4. **Migration/integration:** products, suppliers, customers, balances and machines/interfaces.
5. **Testing/training:** unit, integration, user acceptance, security and role-based training.
6. **Go-live:** phased or big-bang cutover with support desk and contingency plan.
7. **Change management:** apply Kotter; communicate, remove barriers and produce quick wins.
8. **Evaluation:** inventory accuracy, order lead time, closing time, defects, user adoption and ROI.

Class-note case pattern: an integrated SAP ERP allows a sale to update inventory, trigger dispatch, update accounts receivable and appear on dashboards immediately; outcomes include faster month-end closing, higher inventory accuracy and shorter fulfilment lead time.

4.6 NPV Calculation

Definition

Net Present Value discounts future net benefits to today's value and subtracts the initial investment. Positive NPV means the project creates financial value at the chosen discount rate.

Formula / Method

$$NPV = \sum_{t=1}^n \frac{B_t}{(1+r)^t} - C_0$$

B_t = net benefit in year t ; r = discount rate; C_0 = initial cost.

Example: initial cost 8,000,000 BDT; annual net benefit 2,500,000 BDT for 5 years; $r = 10\%$.

$$PV = \frac{2.5m}{1.1} + \frac{2.5m}{1.1^2} + \dots + \frac{2.5m}{1.1^5} \approx 9,476,967$$

$$NPV \approx 9,476,967 - 8,000,000 = 1,476,967 \text{ BDT} > 0$$

Decision: financially acceptable, subject to operational, technical, schedule and risk factors.

4.7 Balanced Scorecard

Definition

The Balanced Scorecard links strategy to measures in four perspectives, preventing managers from judging performance only by short-term finance.

Perspective	Question	Example measures	MIS source
Financial	How do shareholders see us?	Revenue growth, margin, ROI, cost	AIS, budgets, financial MIS
Customer	How do customers see us?	Satisfaction, retention, delivery, market share	CRM, surveys, service system
Internal process	What must we excel at?	Cycle time, defects, fulfilment time, innovation	Production, quality and workflow IS
Learning and growth	Can we continue improving?	Skills, training, system capability, knowledge sharing	HRIS, LMS, KMS, IT metrics

Mnemonic: **Finance - Customer - Internal - Learning (FCIL)**.

4.8 GDSS and EIS Features

GDSS features: electronic brainstorming, simultaneous input, anonymous participation, idea clustering, voting/ranking, shared models, automatic documentation and geographic participation. Benefits include reduced groupthink, equal voice, speed and better traceability.

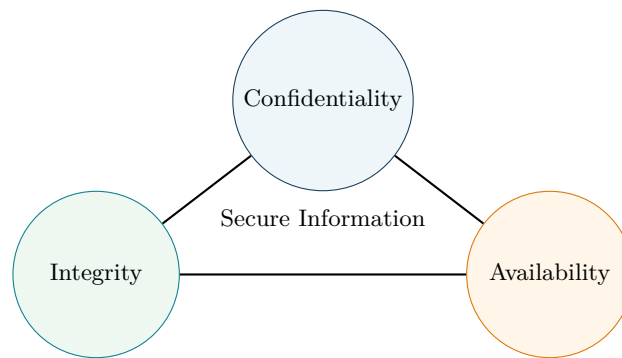
EIS features: graphical dashboards, drill-down, exception highlighting, trend analysis, critical success factors/KPIs, external data, customization, ad-hoc exploration, mobile access and very simple interface.

4.9 EIS Performance Monitoring

1. Define strategic objectives and critical success factors.
2. Select few meaningful KPIs and thresholds.
3. Integrate internal ERP/MIS and external market/regulatory data.
4. Display current value, target, variance and trend.
5. Highlight exceptions using traffic-light status.
6. Allow drill-down from organization to division, product and transaction.
7. Assign owner and corrective action; monitor after action.

Example: executive sees NPA ratio above target, drills from bank total to division and then accounts, assigns recovery action and tracks trend.

4.10 CIA Triad



- **Confidentiality:** only authorized people can access information. Controls: encryption, access control, least privilege.
- **Integrity:** information remains accurate, complete and unaltered. Controls: hashes, validation, digital signatures, audit logs.
- **Availability:** authorized users can access systems when needed. Controls: redundancy, backups, failover, DDoS protection, disaster recovery.

4.11 Simon's Decision-Making Model

1. **Intelligence:** identify and define the problem/opportunity; MIS exceptions and environmental scanning.
2. **Design:** create and analyze alternatives; DSS simulation and what-if analysis.
3. **Choice:** select the best alternative; optimization, scoring and sensitivity analysis.
4. **Implementation:** execute and monitor; TPS performs changes and MIS tracks results.

Mnemonic: **I Design Choices, Implement (IDCI).**

4.12 Anthony's Triangle

Level	Activity	Information/decision	IS support
Top	Strategic planning	Long-term, aggregate, external, unstructured	EIS/ESS, DSS
Middle	Management control	Medium-term, summarized, semi-structured	MIS, DSS
Lower	Operational control	Daily, detailed, real-time, structured	TPS, operational MIS

4.13 Decision Types by Structure

Type	Nature	Example	Main IS
Structured	Routine, clear rules, repetitive, automatable	Reorder stock below threshold	TPS/MIS
Semi-structured	Part model/rule, part judgment	Set price or marketing budget	DSS/MIS
Unstructured	Novel, uncertain, no fixed procedure	Enter a new country or acquire a firm	EIS/DSS

5 Lecture-Only Rescue Topics

These topics are not all on the suggestion sheet, but they are strongly represented in the class notes and the supplied 2024 question paper.

5.1 E-Commerce vs E-Business

E-commerce focuses on online exchange of value; e-business includes all digitally enabled internal and external processes such as ERP, HR, supplier portals and collaboration. **All e-commerce is e-business, but not all e-business is e-commerce.**

5.2 Importance, Opportunities and Challenges of E-Commerce in Bangladesh

Importance/opportunities: 24/7 access, national/global reach, low entry cost, mobile-first consumers, MFS, social commerce, SME market access, data-driven marketing, employment and digital-service export.

Challenges: trust and fraud, COD dominance, weak addressing/last mile, rural and gender digital divide, return handling, payment disputes, cyber risk, inconsistent service, consumer protection, skills and cross-border customs.

5.3 How Encryption Protects Information

Encryption transforms plaintext into unreadable ciphertext using a key. Symmetric encryption such as AES is fast for bulk data; asymmetric encryption such as RSA uses public/private keys for key exchange and identity. Hashing verifies integrity; digital signatures provide integrity, authenticity and non-repudiation; TLS combines these to protect online sessions.

5.4 E-Marketing, Social Media and E-Mail Marketing

E-marketing uses digital channels to attract, convert and retain customers. Benefits: global reach, precise targeting, measurability, personalization, low marginal cost, rapid optimization and two-way engagement.

For a digital-accessories business, social media creates discovery, demonstration, UGC, influencer reach and retargeting. Email supports welcome sequences, product education, abandoned-cart recovery, cross-selling and win-back. Risks are ad dependency, negative comments, spam and privacy; use segmentation, frequency control and clear KPIs.

5.5 Smart Card, E-Cash, E-Banking, E-Trading and E-Finance

- **Smart card:** card containing a chip that stores/processes data and supports secure authentication/payment.
- **E-cash:** digitally represented monetary value used for electronic payments; improves speed and lowers cash handling but requires security, acceptance and regulation.
- **E-banking:** electronic access to banking services such as transfer, balance, bill payment and account management.
- **E-trading:** electronic buying/selling of securities through online platforms.
- **E-finance:** broad use of digital technology in banking, payments, lending, investment, insurance and financial management.

5.6 E-Commerce Security Risks

Phishing/social engineering, malware, payment fraud, credential stuffing, SQL injection, XSS, CSRF, DDoS, insider threats, data leakage, insecure APIs and weak third parties. Use defense-in-depth: secure coding, MFA, encryption, tokenization, monitoring, backups, least privilege, training and incident response.

5.7 Copyright, Plagiarism and Defamation

- **Copyright:** legal protection for original expression; owners control copying, distribution and adaptation.
- **Plagiarism:** presenting another person's words/ideas as one's own without proper acknowledgement; an ethical/academic violation, even when copyright may not be infringed.
- **Defamation:** false statement presented as fact that harms reputation; online posts/reviews may create liability.

5.8 Complementary Assets and Business Processes

Complementary assets are organizational, managerial and social investments needed to realize value from technology: trained people, redesigned processes, supportive culture, standards, partnerships and regulation. Technology alone rarely creates performance.

A **business process** is a logically related set of activities that produces a business result. TPS supports routine execution; MIS supports monitoring/control; DSS supports semi-structured improvement decisions; EIS supports strategic redesign.

5.9 RFID, WSN and Telecommunications Networks

RFID uses radio tags/readers for automatic identification without line-of-sight; it improves inventory accuracy, tracking and checkout. A wireless sensor network consists of distributed sensors that measure and communicate environmental/operational conditions such as temperature, location and vibration. Business value: real-time visibility, predictive maintenance, cold-chain quality and automation.

Principal telecom components: sender/receiver devices, processors/interfaces, channels/media, network software/protocols and communication services. Key technologies include packet switching, TCP/IP, routers/switches, Ethernet/Wi-Fi, cellular, fiber, cloud networking and the Internet.

5.10 Avoiding Redundancy Between IS and IT Functions

Clarify governance and RACI roles; maintain one enterprise architecture and project portfolio; use shared service catalog and CMDB; integrate business analysts with technical teams; eliminate duplicate tools through standards; establish data ownership; conduct joint budgeting and architecture review; measure outcomes rather than departmental territory.

6 Definition Bank

E-Commerce: buying and selling goods, services or information over electronic networks.

E-Business: broader digital enablement of all business processes.

Business model: logic of value creation, delivery and capture.

Revenue model: method by which a firm earns income.

F-Commerce: commerce conducted mainly through Facebook tools.

Multi-tier architecture: separation of application responsibilities into layers.

CDN: geographically distributed servers delivering cached content from near the user.

Scalability: ability to handle increasing work by adding resources.

Conversion rate: percentage of visitors who complete the target purchase.

CAC: average marketing cost to acquire one new customer.

CLV: expected total customer revenue/value over the relationship.

AOV: average revenue per order.

AIDA: Awareness, Interest, Desire, Action marketing funnel.

Fulfilment: storing, picking, packing and delivering orders.

Dropshipping: selling without holding stock; supplier ships to customer.

3PL: external provider handling logistics activities.

4PL: integrator managing the entire logistics network.

EOQ: order quantity minimizing ordering plus holding costs.

Reverse logistics: flow of returned goods from customer back to seller/manufacturer.

Payment gateway: secure interface that captures/encrypts payment data.

Payment processor: routes payment authorization and settlement messages.

Acquirer: merchant's bank/payment institution receiving settlement.

Issuer: bank that issued the customer's card and authorizes it.

Encryption: conversion of plaintext to ciphertext using a key.

Hash: one-way fixed-length digest used to verify integrity.

Digital signature: private-key signature verified by public key.

Network effect: service value increases with number of users.

Metcalfe's Law: network value is approximately proportional to n^2 .

Winner-takes-all: market concentration caused by reinforcing platform advantages.

RFM: Recency-Frequency-Monetary customer segmentation.

MIS: integrated people-process-data-technology system providing management information.

TPS: system processing routine high-volume transactions reliably.

DSS: interactive system supporting semi-structured decisions with models/data.

EIS: executive system providing visual strategic information and exceptions.

MRS: system generating scheduled, exception and demand reports.

DPS: system focused on efficient data processing and records.

ERP: integrated enterprise platform and database across functions.

DIKW: Data-Information-Knowledge-Wisdom hierarchy.

KPI: measurable indicator of progress toward an objective.

Balanced Scorecard: performance framework with four strategic perspectives.

NPV: present value of future net benefits minus initial cost.

Expected value: probability-weighted average payoff.

Structured decision: routine decision with clear rules.

Semi-structured decision: partly model-based and partly judgmental.

Unstructured decision: novel decision without a defined procedure.

GDSS: system supporting collaborative group decision activities.

CIA Triad: confidentiality, integrity and availability objectives.

Complementary assets: non-technology investments required for IT value.

Business process: related activities producing a defined business result.

RFID: radio-frequency identification using tags and readers.

WSN: distributed wireless sensors that measure and communicate conditions.

7 Difference Bank

Comparison	First concept	Second concept
E-Commerce vs E-Business	Online commercial transactions/exchange of value	All digitally enabled business processes; broader
Business model vs Revenue model	How value is created/delivered/captured	Specifically how money is earned
B2B vs B2C	Business sells to business; larger, negotiated, relationship-based	Business sells to individual consumers; high volume, standardized
Subscription vs Freemium	Most users pay recurring fee for access	Basic tier free; minority convert to premium
E-Commerce vs F-Commerce	Any digital channel/platform	Facebook-based subset with social interaction
Gateway vs Processor	Captures/encrypts payment data at checkout	Routes transaction through financial networks
Issuer vs Acquirer	Customer's bank; authorizes and bears cardholder risk	Merchant's bank; accepts and settles transactions
Authorization vs Settlement	Real-time approval/reservation of funds	Later transfer and credit of actual funds
Symmetric vs Asymmetric encryption	Same secret key; fast; bulk data	Public/private keys; slower; exchange/signatures
Hashing vs Encryption	One-way integrity digest; not designed to reverse	Reversible with correct key; confidentiality
In-house vs Dropshipping	Own inventory and fulfilment; control/high cost	No inventory; low capital/low control
3PL vs 4PL	Executes logistics functions	Coordinates total supply chain and multiple providers
SEO vs SEM	Organic ranking, no fee per click, slower accumulation	Paid search advertising, immediate traffic, CPC
CR vs CTR	Purchase completion divided by visits	Ad clicks divided by impressions
CAC vs CLV	Cost to acquire a customer	Value/revenue expected from the customer
TPS vs MIS	Detailed live transactions, operational users	Summaries/reports, middle management
MIS vs DSS	Predefined reports; what happened	Interactive models; what could/should happen
DSS vs EIS	Analytical support for semi-structured decisions	High-level monitoring for executives/unstructured decisions
Data vs Information	Raw facts without context	Processed facts with meaning/context
Information vs Knowledge	Contextual answer to what/when/where	Internalized pattern/rule explaining how
IS vs IT	People + process + data + technology for business purpose	Technology tools/infrastructure only; subset of IS
ERP vs MIS	Enterprise transaction/process integration platform	Broader management-information concept; ERP can feed MIS
Structured vs Unstructured decision	Clear procedure and repeatability	Novel, uncertain and judgment-dominant
Preventive vs Detective control	Stops event before it happens	Finds event during/after occurrence
Copyright vs Plagiarism	Legal right over original expression	Ethical/academic misrepresentation of authorship
RFID vs Barcode	Radio, no line-of-sight, multiple tags, higher cost	Optical, line-of-sight, one-at-a-time, low cost

8 Formula and Mathematics Bank

Formula / Method

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$ROP = dL + SS$$

$$NPV = \sum_{t=1}^n \frac{B_t}{(1+r)^t} - C_0$$

$$CR = \frac{\text{Purchases}}{\text{Visits}} \times 100\%$$

$$CAC = \frac{\text{Marketing Spend}}{\text{New Customers}}$$

$$CLV = AOV \times \text{Frequency} \times \text{Lifespan}$$

$$\text{Bounce} = \frac{\text{SinglePageVisits}}{\text{Visits}} \times 100\%$$

$$EF = ES + d$$

$$\text{Float} = LS - ES = LF - EF$$

$$TC(Q) = \frac{D}{Q}S + \frac{Q}{2}H$$

$$EV = \sum P_i V_i$$

$$V_{\text{network}} \propto n^2$$

$$CPC = \frac{\text{Ad Spend}}{\text{Clicks}}$$

$$AOV = \frac{\text{Revenue}}{\text{Orders}}$$

$$ROAS = \frac{\text{Ad Revenue}}{\text{Ad Spend}}$$

$$\text{Abandonment} = \left(1 - \frac{\text{Orders}}{\text{Carts}}\right) 100\%$$

$$LS = LF - d$$

$$\text{Links}_{\text{mesh}} = \frac{n(n-1)}{2}$$

Four calculation checks

1. **CPM:** project duration equals EF of final activity; all critical floats are zero.
2. **EOQ:** at optimum, annual ordering and holding costs should be approximately equal.
3. **EV:** probabilities for each alternative must sum to 1.
4. **NPV:** future positive cash flow must be discounted downward; do not add initial cost again.

Practice answers

1. 2,000 visits and 60 purchases: $CR = 3\%$.
2. 40,000 BDT ad spend and 2,000 clicks: $CPC = 20$ BDT.
3. 300 carts and 90 orders: $\text{abandonment} = (1 - 90/300)100 = 70\%$.
4. Network users rise from 100 to 300: approximate value multiplier $= (300/100)^2 = 9$.
5. Full mesh of 8 nodes: $8 \times 7/2 = 28$ links.

9 Mnemonic Bank

Topic	Mnemonic
AIDA	Awareness, Interest, Desire, Action - “Attract, Inform, Develop want, Ask”
Value-chain primary	“I Operate Outlets, Market, Serve”
Value-chain support	“Firm Humans Procure Technology” (Infrastructure, HR, Procurement, Technology)
CPM	“Forward, Backward, Float, Critical”
EOQ	“Double D and S, divide H, root”
RFM	“Recent, Regular, Revenue”
Kotter	“Urgent Coalitions Visualize Volunteers; Enable Wins, Sustain, Institute”
DIKW	“Facts, Meaning, Pattern, Judgment”
Simon	“I Design Choices, Implement”
Anthony	“Strategic - Management - Operational”
CIA	“Private, Correct, Present” = Confidential, intact, available
Balanced Scorecard	“Finance, Customer, Internal, Learning”
Card participants	“Customer Merchant Gateway Processor Network Issuer; Acquirer pays merchant”
Revenue models	“Ads, Subscribe, Transact, Free-Premium, Affiliate, Sell”
IS hierarchy	“Transactions - Reports - Decisions - Executives”
Information quality	“Accurate, Complete, Relevant, Timely, Reliable, Accessible”
Security controls	“Prevent, Detect, Correct, Recover”
ERP implementation	“Plan, Map, Configure, Migrate, Test, Train, Go-live, Improve”

10 Quick Keyword Bank

Question area	Keywords that should appear
Architecture	presentation, business logic, data, separation, scalability, security, API, load balancing
Value chain	five primary, four support, value/cost, competitive advantage, e-commerce application
AIDA	customer stage, tactic, metric, funnel drop-off, conversion
EOQ	ordering cost, holding cost, optimal quantity, assumptions, reorder point
RFM	recency, frequency, monetary, 1–5 scoring, segment, action
Payment flow	gateway, processor, network, issuer, authorization, clearing, settlement, acquirer
MFS fraud	PIN/OTP, vishing, phishing, fake SMS, SIM swap, verification, anomaly detection
MIS hierarchy	operational/tactical/strategic, detail/aggregation, TPS-MIS-DSS-EIS
ERP	integration, unified database, modules, real-time, silos, change management, data migration
Kotter	urgency, coalition, vision, volunteers, barriers, wins, acceleration, institutionalize
BSC	financial, customer, internal process, learning/growth, KPI, strategy link
EIS	dashboard, drill-down, exception, trend, external data, CSF, customization
CIA	confidentiality-access control, integrity-hash/signature, availability-backup/redundancy
Decision	structured, semi-structured, unstructured; Simon intelligence-design-choice-implementation
Network effects	direct, indirect/two-sided, data, tipping point, lock-in, multi-homing, Metcalfe

11 Likely Question Skeletons

1. **Find the critical path.** Draw network; forward pass; backward pass; table; zero float; state duration.
2. **Explain Porter's Value Chain in e-commerce.** Definition; 5 primary; 4 support; labelled diagram; platform example; competitive benefit.
3. **Apply AIDA to an online accessories firm.** Define four stages; tactic and metric at each; short applied campaign.
4. **Calculate EOQ.** Formula/variables; substitution; round; orders/year; interpretation; assumptions.
5. **Use RFM to set marketing strategy.** Define/scoring; classify customers; choose action for each segment.
6. **Explain Kotter for ERP implementation.** Eight steps tied to resistance, training, pilot wins and institutionalization.
7. **Calculate expected value and decide.** Probability table; weighted payoffs; compare; risk caveat.
8. **Compare TPS, MIS, DSS and EIS.** Management level, decision type, data, output, horizon and example.
9. **ERP in manufacturing.** Pre-problem; modules/integration; implementation steps; benefits; risks; KPIs.
10. **Card processing flow.** Participants; authorization; clearing/settlement; security; distinction among gateway/processor/acquirer.
11. **Discuss e-commerce opportunities/challenges in Bangladesh.** Balanced headings; MFS/social commerce; trust, logistics, payment, divide and regulation.
12. **Explain CIA and security controls.** Define each; attack example; preventive control; e-commerce/MFS application.

Exam Tip

For every 5-mark framework question, draw the model first. A correct labelled diagram creates structure, saves writing time and makes the answer easy to mark.

12 Final 30-Minute Revision

1. Recite seven starred-topic mnemonics without looking.
2. Write CPM, EOQ, EV, NPV and all e-commerce metric formulas once.
3. Draw Value Chain, AIDA, DIKW, IS hierarchy, Anthony triangle and CIA from memory.
4. Say the card flow aloud: customer → merchant → gateway → processor → network → issuer; response back; settlement to acquirer/merchant.
5. Review Difference Bank: MIS vs DSS, TPS vs MIS, subscription vs freemium, gateway vs processor, issuer vs acquirer.
6. Sleep enough to preserve recall; do not begin an entirely new long topic in the final minutes.

You do not need perfect memory. You need clear structure, correct keywords, labelled diagrams and accurate mathematics.